

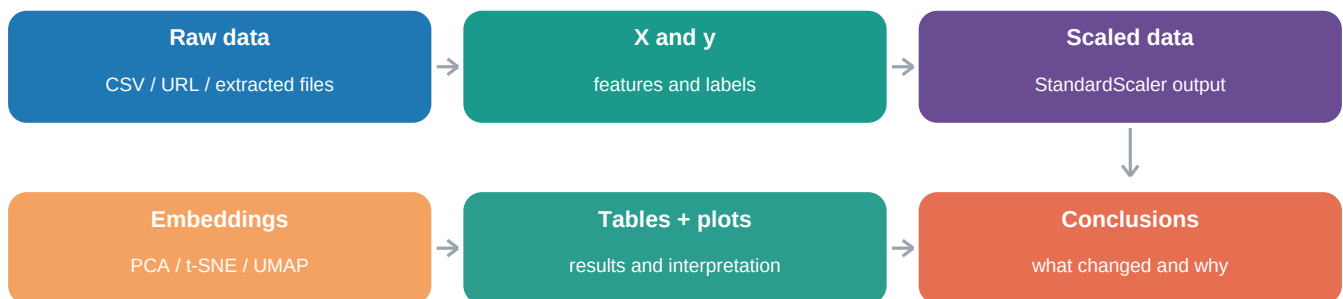
Day 5 PCA

Variable Flow Guide

A practical map of how variables move through the Wine, RNA-Seq, and Expert PCA notebooks

Purpose of this document

This guide focuses on variable flow: what each variable stores, where it comes from, its shape, and where it is used next. It is designed as a quick debugging and explanation companion for the three Day 5 PCA notebooks.



Prepared for Ruturaj Mokashi

1. How to Read This Guide

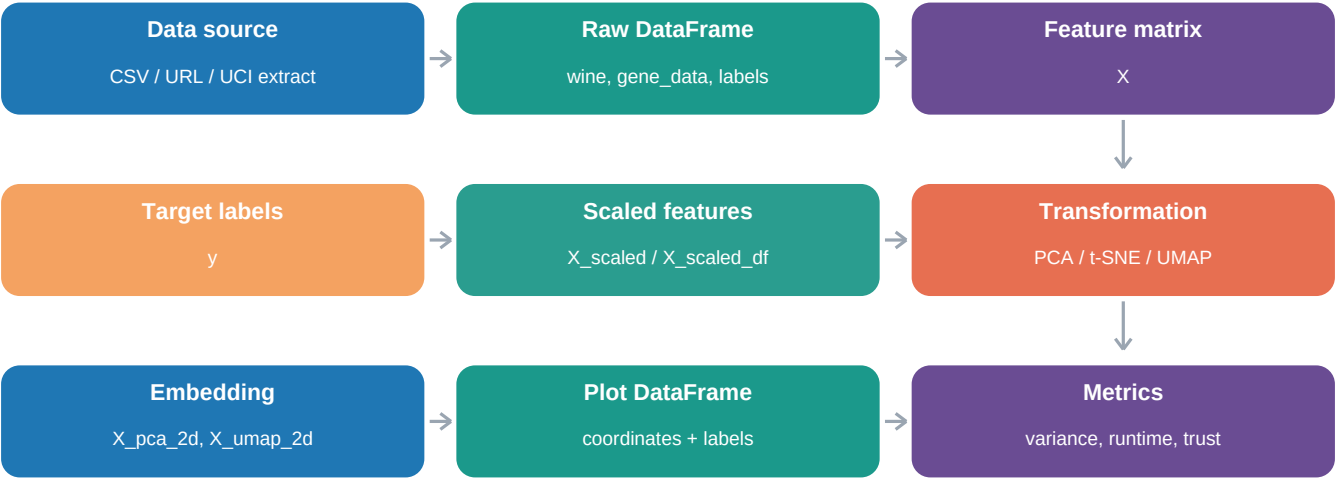
Each notebook has the same basic pattern: raw data is loaded, separated into features and labels, scaled, transformed, and finally used for visualisation or modelling. The most important part is understanding what each variable contains at each step.

Symbol	Simple meaning	Example
x	Input features only. This is what PCA, t-SNE, UMAP, or models use.	Wine chemistry columns; gene expression columns
y	Target or label column. Not used by unsupervised embeddings, except for colouring plots.	quality; tumour Class
scaled	Standardised version of X. Features are placed on comparable scale.	X_scaled, X_scaled_df
pca / tsne / umap	Fitted transformation object or produced embedding.	pca_2d, tsne, umap_reducer
df	A DataFrame used for clean display, plotting, or summary tables.	pca_2d_df, runtime_df
runtime / trust	Evaluation variables that store speed or quality metrics.	pca_runtime, tsne_trust

Key rule

Position variables such as PC1/PC2 come from feature data only. Colour variables such as quality or tumour class are added later for interpretation. This is why PCA, t-SNE, and UMAP remain unsupervised in these notebooks.

2. Global Variable Flow Pattern

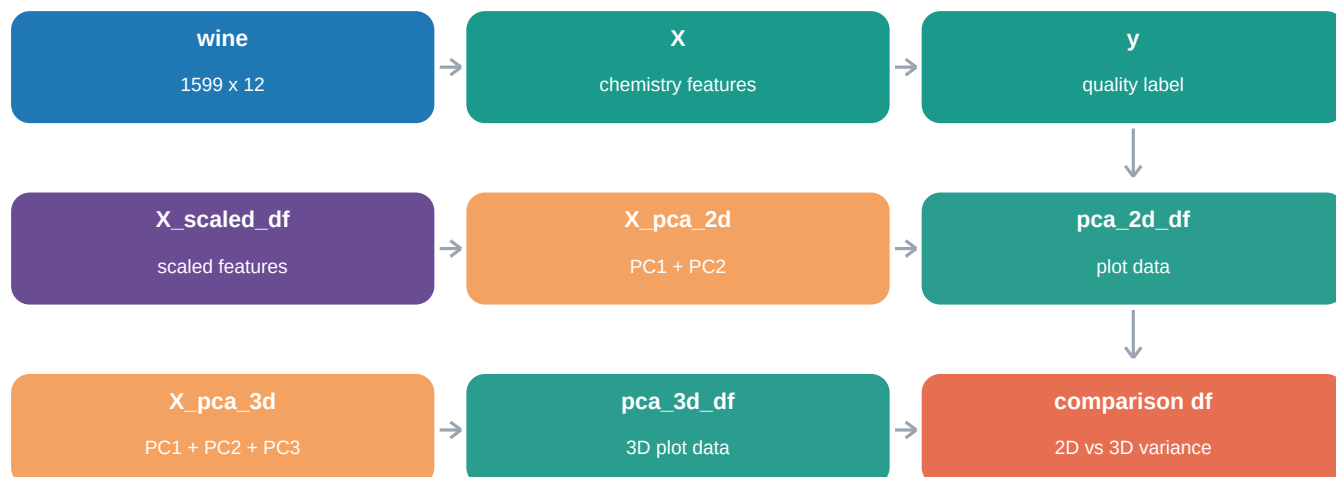


This pattern repeats across all three notebooks. The names change slightly, but the logic is the same.

Stage	Question to ask	Common variables
Load	What did we read from disk or URL?	wine, gene_data, labels
Separate	Which columns are inputs and which are labels?	X, y, sample_ids
Scale	What data is ready for PCA/t-SNE/UMAP?	X_scaled_df, X_scaled
Transform	Which method created a lower-dimensional representation?	pca_2d, pca_full, tsne, umap_reducer
Store	Where are the transformed coordinates saved?	X_pca_2d, X_tsne_2d, X_umap_2d
Evaluate	How much variance, how fast, or how trustworthy?	explained_variance_ratio, runtime_df, trustworthiness_df

3. Notebook 1 - Wine Quality PCA Variable Flow

Goal: reduce 11 wine chemistry features into 2D and 3D PCA views, then colour the plots by wine quality.



Variable	Shape / type	Created from	Purpose and next use
wine	DataFrame (1599, 12)	pd.read_csv(url, sep=';')	Raw Wine Quality dataset. Contains 11 input columns plus quality.
X	DataFrame (1599, 11)	wine.drop(columns=['quality'])	Input chemistry features used by PCA.
y	Series (1599,)	wine['quality']	Wine quality labels. Used only for colouring scatter plots.
scaler	StandardScaler object	StandardScaler()	Learns mean and standard deviation of each input feature.
X_scaled	NumPy array (1599, 11)	scaler.fit_transform(X)	Scaled input values for PCA.
X_scaled_df	DataFrame (1599, 11)	pd.DataFrame(X_scaled, columns=X.columns)	Readable scaled feature matrix.
pca_2d	PCA object	PCA(n_components=2)	Learns the first two PCA directions.
X_pca_2d	Array (1599, 2)	pca_2d.fit_transform(X_scaled_df)	2D PCA coordinates: PC1 and PC2.
pca_2d_df	DataFrame (1599, 3)	PC1, PC2 + y	Used for the 2D scatter plot coloured by quality.
pca_3d	PCA object	PCA(n_components=3)	Learns first three PCA directions.
X_pca_3d	Array (1599, 3)	pca_3d.fit_transform(X_scaled_df)	3D PCA coordinates: PC1, PC2, PC3.
pca_3d_df	DataFrame (1599, 4)	PC1, PC2, PC3 + y	Used for the 3D scatter plot coloured by quality.

Notebook 1 - Results Variables

Result variable / value	Meaning	Interpretation
pca_2d.explained_variance_ratio_	PC1 = 28.17%, PC2 = 17.51%	The first two components keep 45.68% of wine feature variance.
pca_3d.explained_variance_ratio_	PC1 + PC2 + PC3 = 59.78%	The 3D view keeps more information than the 2D view.
pca_comparison_df	2D vs 3D variance table	Used to compare how much information each visualisation keeps.

Do not confuse

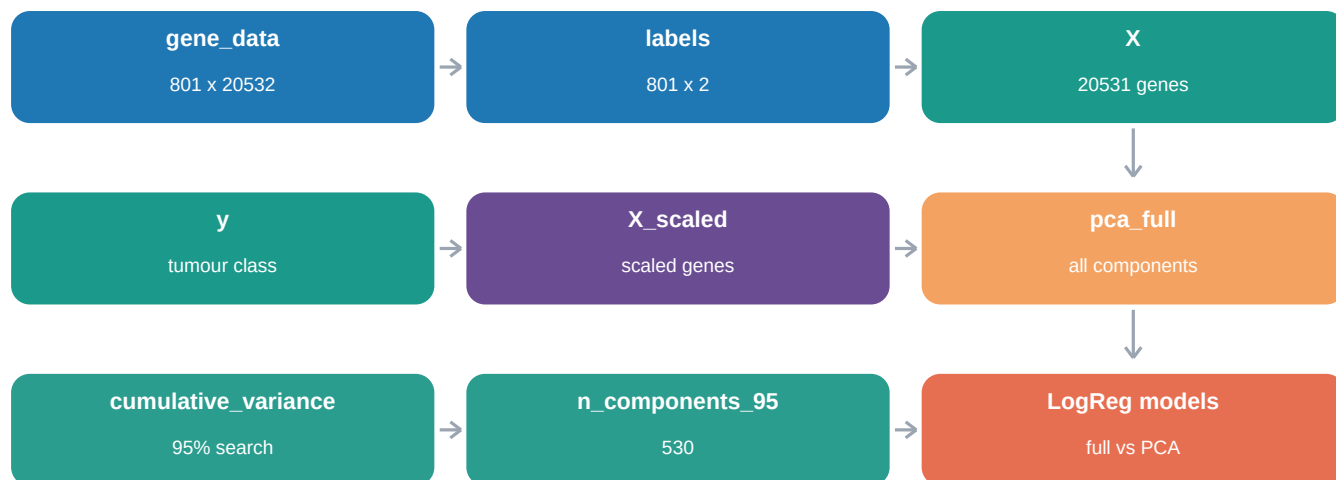
quality is never used to create PC1, PC2, or PC3. It is added later as a colour variable to visually inspect whether the PCA space shows quality patterns.

Core code pattern

```
X = wine.drop(columns=['quality'])
y = wine['quality']
X_scaled_df = pd.DataFrame(StandardScaler().fit_transform(X), columns=X.columns)
X_pca_2d = PCA(n_components=2).fit_transform(X_scaled_df)
```

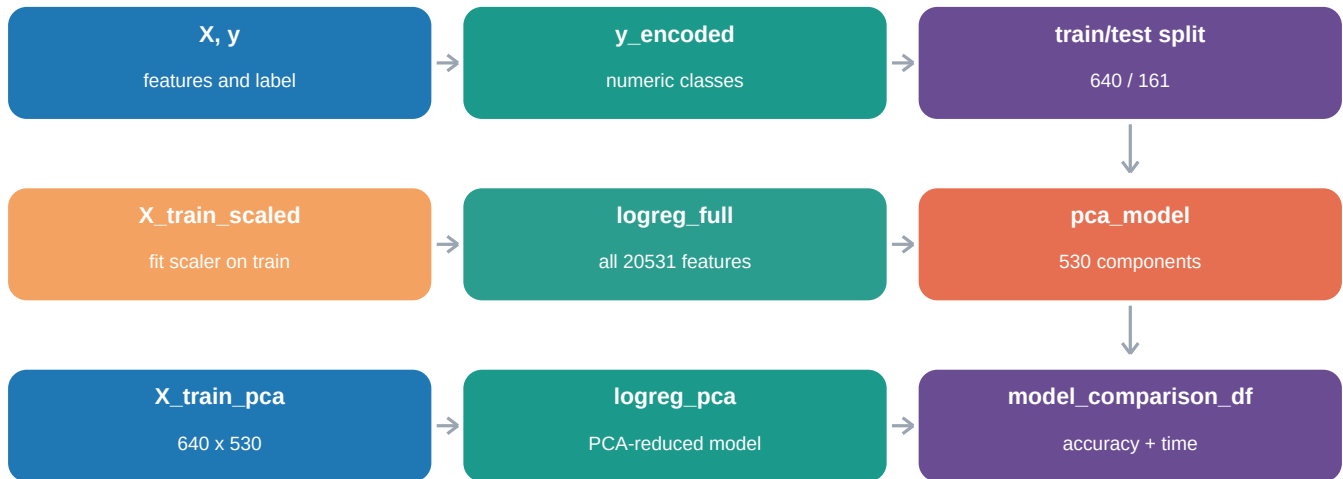
4. Notebook 2 - RNA-Seq PCA + Logistic Regression Variable Flow

Goal: reduce 20,531 gene features using PCA, find 95% variance components, train Logistic Regression, and inspect loadings.



Variable	Shape / type	Created from	Purpose and next use
gene_data	DataFrame (801, 20532)	pd.read_csv(data.csv)	Raw gene expression file. Includes sample ID plus 20,531 genes.
labels	DataFrame (801, 2)	pd.read_csv(labels.csv)	Sample ID and tumour class labels.
sample_ids	Series (801,)	gene_data['Unnamed: 0']	Reference IDs. Not used as features.
X	DataFrame (801, 20531)	gene_data.drop('Unnamed: 0')	All gene expression input features.
y	Series (801,)	labels['Class']	Tumour type labels.
gene_names	List length 20531	X.columns.tolist()	Used to label PCA loadings by gene name.
X_scaled	Array (801, 20531)	StandardScaler().fit_transform(X)	Scaled gene matrix for PCA.
pca_full	PCA object	PCA()	Learns all possible PCA components.
X_pca_full	Array (801, 801)	pca_full.fit_transform(X_scaled)	Full PCA-transformed representation.
explained_variance_ratio	Array (801,)	pca_full.explained_variance_ratio_	Variance explained by each component.
cumulative_variance	Array (801,)	np.cumsum(explained_variance_ratio)	Used to find how many components reach 95%.
n_components_95	Integer: 530	argmax(cumulative_variance >= 0.95) + 1	Number of PCA components needed for 95% variance.

Notebook 2 - Modelling Variable Flow



Variable	Shape / type	Created from	Purpose and next use
label_encoder	LabelEncoder object	LabelEncoder()	Converts class labels such as BRCA/KIRC into numeric labels.
y_encoded	Array (801,)	label_encoder.fit_transform(y)	Numeric target for Logistic Regression.
X_train, X_test	(640, 20531), (161, 20531)	train_test_split(X, y_encoded)	Train/test feature sets before scaling.
y_train, y_test	(640,), (161,)	train_test_split	Train/test labels.
scaler_model	StandardScaler object	StandardScaler()	Fitted only on training data to avoid leakage.
X_train_scaled	Array (640, 20531)	scaler_model.fit_transform(X_train)	Scaled training data.
X_test_scaled	Array (161, 20531)	scaler_model.transform(X_test)	Scaled test data using train scaler.
logreg_full	LogisticRegression	fit on X_train_scaled	Full-feature model.
pca_model	PCA(530)	PCA(n_components=n_components_95)	Learns 530-component reduction on training data.
X_train_pca	Array (640, 530)	pca_model.fit_transform(X_train_scaled)	PCA-reduced training features.
X_test_pca	Array (161, 530)	pca_model.transform(X_test_scaled)	PCA-reduced test features.
logreg_pca	LogisticRegression	fit on X_train_pca	PCA-reduced model.
model_comparison_df	DataFrame	accuracy + runtime values	Compares full vs PCA model.

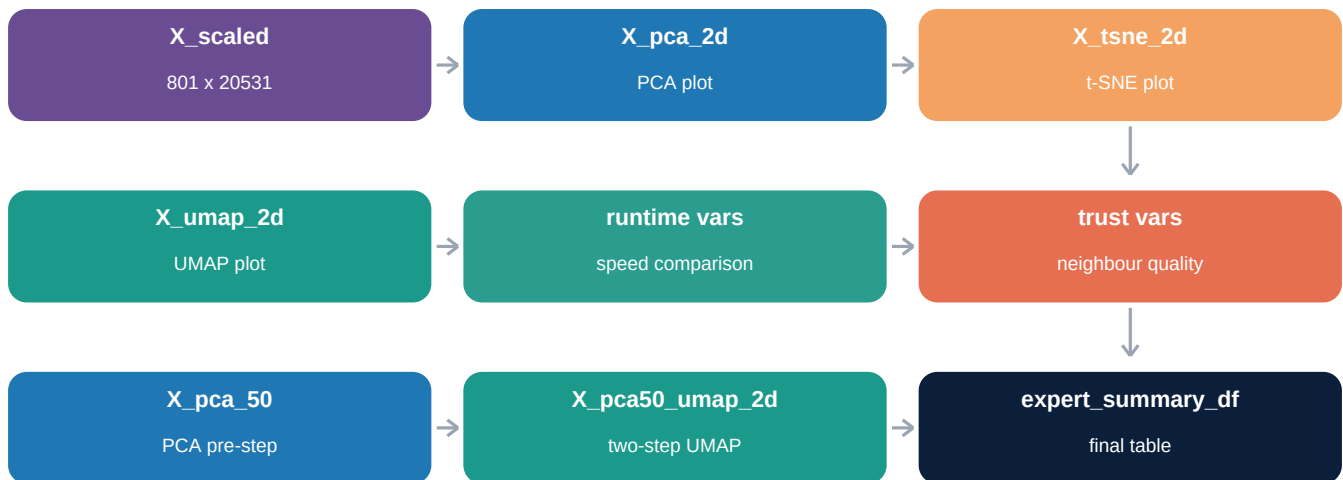
Notebook 2 - Loading Interpretation Variables

Variable	Shape / type	Created from	Purpose and next use
pc1_loadings	Array (20531,)	pca_full.components_[0]	Feature weights for PC1.
pc2_loadings	Array (20531,)	pca_full.components_[1]	Feature weights for PC2.
loadings_df	DataFrame (20531, 4)	gene_names + PC1/PC2 loadings	Main table for interpreting gene contributions.
Absolute PC1 Loading	Column	abs(PC1 Loading)	Contribution strength regardless of sign.
top_pc1_genes	DataFrame (20, 4)	loadings_df sorted by Absolute PC1 Loading	Top genes driving PC1.
heatmap_data	DataFrame (20, 2)	top_pc1_genes indexed by Gene	Used for PC1/PC2 loading heatmap.

Notebook 2 key results
20,531 gene features were reduced to 530 PCA components for 95.01% variance. Full Logistic Regression accuracy: 98.14%. PCA-reduced Logistic Regression accuracy: 100.00%. Training time dropped from 0.56s to 0.03s after PCA transformation.

5. Notebook 3 - Expert PCA vs t-SNE vs UMAP Variable Flow

Goal: compare PCA, t-SNE, UMAP, and PCA(50)+UMAP embeddings using runtime, trustworthiness, and visual separation.



Variable	Shape / type	Created from	Purpose and next use
X_scaled	Array (801, 20531)	StandardScaler().fit_transform(X)	Reference high-dimensional scaled data for all embeddings.
pca_2d	PCA object	PCA(n_components=2)	Creates linear 2D PCA embedding.
X_pca_2d	Array (801, 2)	pca_2d.fit_transform(X_scaled)	Coordinates for PCA scatter plot and trustworthiness.
pca_runtime	Float seconds	time before/after PCA	Runtime comparison.
tsne	TSNE object	TSNE(n_components=2, perplexity=30)	Creates non-linear t-SNE embedding.
X_tsne_2d	Array (801, 2)	tsne.fit_transform(X_scaled)	Coordinates for t-SNE scatter plot and trustworthiness.
tsne_runtime	Float seconds	time before/after t-SNE	Runtime comparison.
umap_reducer	UMAP object	umap.UMAP(n_neighbors=15)	Creates direct UMAP embedding.
X_umap_2d	Array (801, 2)	umap_reducer.fit_transform(X_scaled)	Coordinates for direct UMAP scatter plot and trustworthiness.
umap_runtime	Float seconds	time before/after UMAP	Runtime comparison.

Notebook 3 - Trustworthiness and Comparison Variables

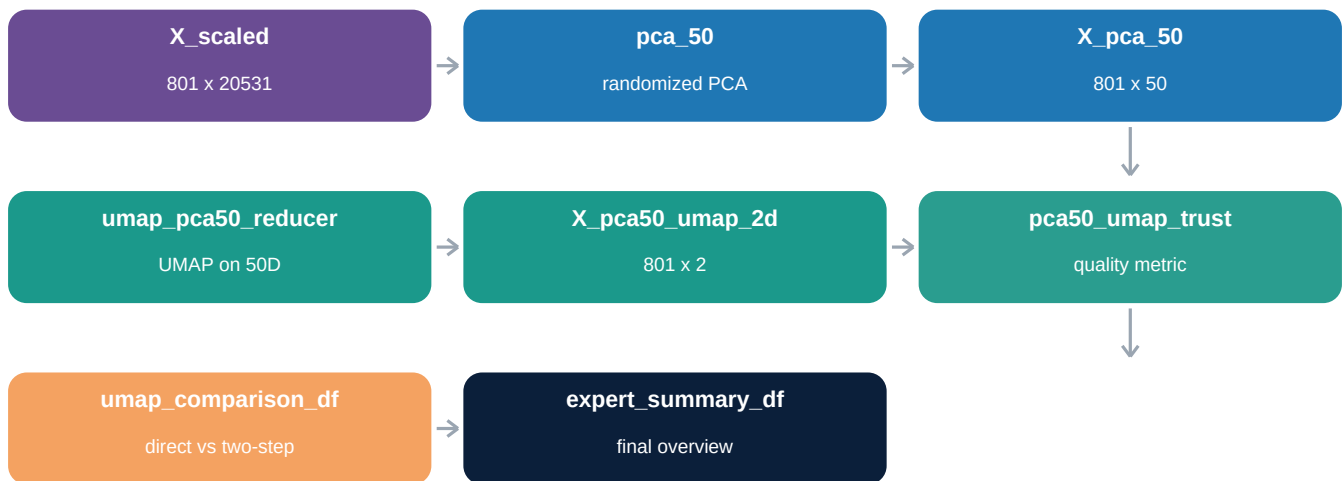
Variable	Shape / type	Created from	Purpose and next use
trust_n_neighbors	Integer: 15	manual setting	Number of neighbours used in trustworthiness metric.
pca_trust	Float: 0.7863	trustworthiness(X_scaled, X_pca_2d)	Measures PCA local-neighbour preservation.
tsne_trust	Float: 0.9415	trustworthiness(X_scaled, X_tsne_2d)	Measures t-SNE local-neighbour preservation.
umap_trust	Float: 0.9183	trustworthiness(X_scaled, X_umap_2d)	Measures UMAP local-neighbour preservation.
trustworthiness_df	DataFrame (3, 2)	method names + trust scores	Comparison table for PCA/t-SNE/UMAP.
runtime_df	DataFrame (3, 2)	method names + runtimes	Comparison table for PCA/t-SNE/UMAP runtime.
pca_plot_df	DataFrame (801, 3)	X_pca_2d + y	PCA coordinates plus tumour labels for plotting.
tsne_plot_df	DataFrame (801, 3)	X_tsne_2d + y	t-SNE coordinates plus tumour labels for plotting.
umap_plot_df	DataFrame (801, 3)	X_umap_2d + y	UMAP coordinates plus tumour labels for plotting.

Trustworthiness interpretation

Higher means the 2D plot better preserves original local neighbourhoods. In the executed notebook outputs:
t-SNE = 0.9415, UMAP = 0.9183, PCA = 0.7863.

Notebook 3 - PCA(50) + UMAP Variable Flow

This section tracks the two-step expert idea: first compress the full gene matrix to 50 PCA components, then run UMAP on the 50-component representation.



Variable	Shape / type	Created from	Purpose and next use
pca_50	PCA object	<code>PCA(n_components=50, svd_solver='randomized')</code>	Compresses full scaled data to 50 PCA components.
X_pca_50	Array (801, 50)	<code>pca_50.fit_transform(X_scaled)</code>	Input to UMAP in the two-step workflow.
pca_50_runtime	Float seconds	timer around PCA(50)	Preprocessing runtime for two-step workflow.
pca_50_variance	Float	<code>pca_50.explained_variance_ratio_.sum()</code>	How much variance the 50 PCA components preserve.
umap_pca50_reducer	UMAP object	<code>umap.UMAP(n_neighbors=15)</code>	UMAP fitted on X_pca_50 instead of X_scaled.
X_pca50_umap_2d	Array (801, 2)	<code>umap_pca50_reducer.fit_transform(X_pca_50)</code>	2D embedding for PCA(50)+UMAP plot.
pca50_umap_runtime	Float seconds	timer around UMAP on X_pca_50	UMAP runtime after PCA compression.
pca50_umap_trust	Float	<code>trustworthiness(X_scaled, X_pca50_umap_2d)</code>	Neighbour preservation score of two-step embedding.
umap_comparison_df	DataFrame	direct UMAP + PCA(50)+UMAP metrics	Compares input features, time, and trustworthiness.
expert_summary_df	DataFrame	all method metrics	Final expert-level summary table.

6. Common Mistakes and How to Debug Variables

Mistake	Symptom	Fix
Using y inside PCA	PCA appears supervised or target leakage risk	Use only X or X_scaled in fit_transform. Add y only after embedding for colour.
Forgetting to drop ID columns	PCA/model uses text column or crashes	Drop Unnamed: 0 before scaling or modelling.
Scaling after PCA	Incorrect PCA directions	Scale X first, then fit PCA/t-SNE/UMAP where relevant.
Fitting scaler on full data before model split	Data leakage in modelling	For modelling, split first, fit scaler only on X_train.
Confusing loadings and scores	Wrong interpretation of PCA axes	Scores = row coordinates. Loadings = feature/gene contributions.
Trustworthiness uses wrong reference	Scores not comparable	Use original high-dimensional X_scaled as first argument.
Interpreting t-SNE/UMAP distances globally	Overclaiming cluster distance meaning	Interpret local neighbourhoods and cluster separation carefully.

Fast check before every plot

Ask: What determines point position? What determines point colour? In these notebooks, position comes from transformed features; colour comes from y labels added afterwards.

7. Final Variable Cheat Sheet

Variable pattern	Meaning	Typical examples
raw data	Data loaded directly from source	wine, gene_data, labels
x	Input features only	wine chemistry features; gene expression columns
y	Target labels only	quality; Class
*_scaled	Standardised feature matrix	X_scaled, X_scaled_df, X_train_scaled
pca_*	PCA model object	pca_2d, pca_full, pca_model, pca_50
X_pca_*	Data transformed by PCA	X_pca_2d, X_pca_full, X_train_pca, X_pca_50
*_plot_df	Coordinates plus labels for plotting	pca_plot_df, tsne_plot_df, umap_plot_df
*_runtime	Execution time in seconds	pca_runtime, tsne_runtime, umap_runtime
*_trust	Trustworthiness score	pca_trust, tsne_trust, umap_trust
*_comparison_df	Summary table	model_comparison_df, umap_comparison_df
loadings_*	Feature contribution interpretation	loadings_df, top_pc1_genes

One-sentence memory

Remember

Raw data becomes X and y. X becomes scaled data. Scaled data becomes PCA/t-SNE/UMAP coordinates. Coordinates plus y become plots. Metrics explain how much information, speed, or neighbourhood structure was preserved.